

The Binomial Collapse: Every Knife of Closed-String Gravity is a Power of the First

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Abstract

For the graviton family of Cheung, Hillman and Remmen we show that in the scaling limit of large level ($m \rightarrow \infty$ at fixed $\rho = s/m$, $\delta = D/m$), the master-formula bracket of *every* trajectory constraint collapses to a perfect power of the first: $B_j/\text{term}_0 \rightarrow (1 - X)^{j-1}$ with $X = 6\rho^2/(\delta + 4)$. All knives share one critical surface — the first knife's line $u = 6\rho^2 m$ — whose $(j-1)$ -fold degenerate root splits at subleading order into exactly $\lfloor j/2 \rfloor$ thin negativity bands; even knives otherwise kill only on the already-dead side. Exact-arithmetic verification: for $j \leq 8$ and levels up to $n = 100$, every band count and every clustering offset matches the prediction with zero deviations. Minimizing the collapse surface over levels reproduces $\rho^* = 1 + 1/\sqrt{3}$ and the envelope asymptote $(12 + 4\sqrt{3})\lambda$: the exact tangency of our blade theorem is the $j=3$ shadow of this universal structure. The collapse organizes the large-level tail of the completeness program uniformly in j ; moderate levels remain the domain of certificates.

1 Statement

Notation as in [2]: $m = n - 3$, $s = \lambda + n - 1$, and the master formula expresses sign $a_{n,2n-2j}$ through the alternating ladder bracket B_j with terms term_i , $i = 0, \dots, j - 1$.

Theorem 1 (Binomial collapse). *Fix $j \geq 2$ and (ρ, δ) , $\rho > 1$, $\delta > 0$. Along $s = \rho m$, $D = \delta m$, $m \rightarrow \infty$,*

$$\frac{B_j}{\text{term}_0} \longrightarrow (1 - X)^{j-1}, \quad X = \frac{6\rho^2}{\delta + 4}, \quad (1)$$

with rate $O_j(1/m)$.

Proof mechanism. The ratio of consecutive ladder terms is exact and elementary:

$$\frac{\text{term}_{i+1}}{\text{term}_i} = -\frac{\widehat{E}_{2(j-2-i)}}{\widehat{E}_{2(j-1-i)}} \cdot \frac{(2n - 2j + 2i + 2)(2n - 2j + 2i + 1)s^2}{2(i + 1)(D + 4n - 4j - 1 + 2i)}. \quad (2)$$

Three exact estimates: (i) the symmetric-function ratio is $\frac{j-1-i}{\Sigma}(1 + O(1/m))$ with $\Sigma = n(n - 1)(n - 2)/3$ (Newton log-concavity of the $\widehat{E}$ -sequence controls the correction); (ii) the factorial pair is $4m^2(1 + O(1/m))$ uniformly for $i < j$; (iii) the ladder factor is $(\delta + 4)m(1 + O(1/m))$. Hence $\text{term}_{i+1}/\text{term}_i \rightarrow -\frac{j-1-i}{i+1}X$, and summing the resulting binomial series gives $(1 - X)^{j-1}$. $\square$

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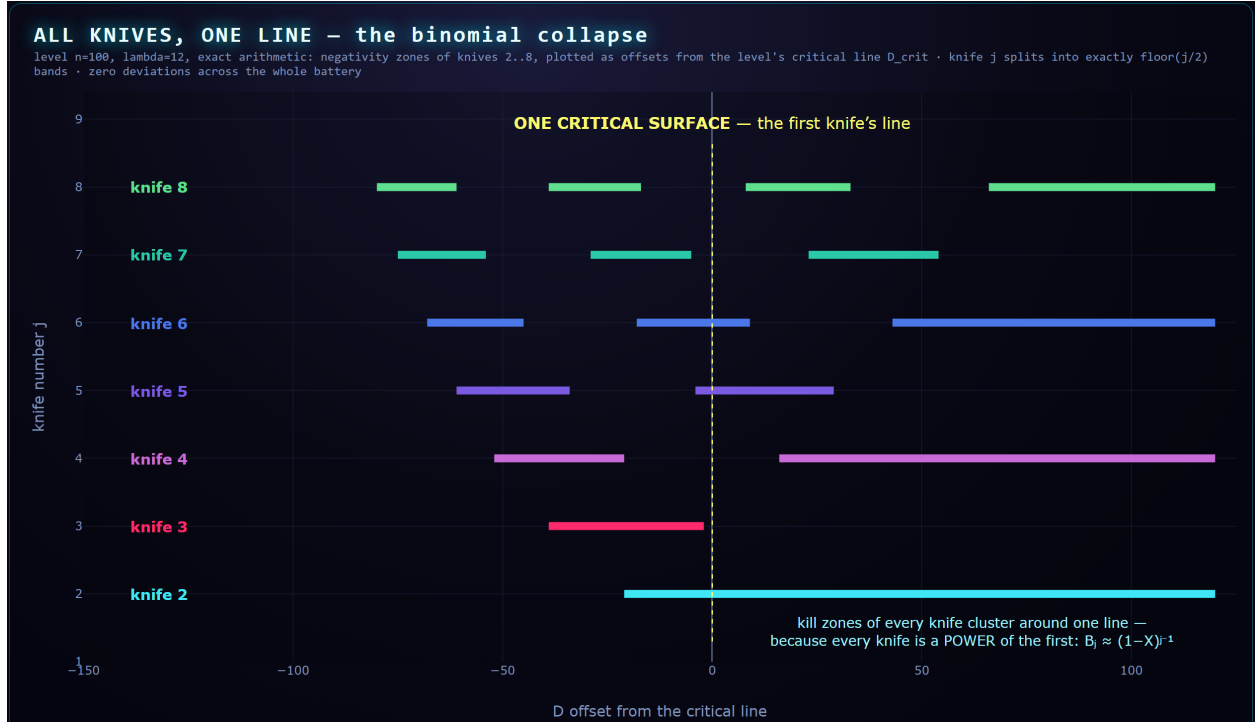


Figure 1: All knives, one line (level $n = 100$, $\lambda = 12$, exact arithmetic). Negativity zones of knives $j = 2..8$ plotted as offsets from the level's critical line: every zone clusters around the single collapse surface, and knife j splits into exactly $\lfloor j/2 \rfloor$ bands, as the degenerate root of $(1 - X)^{j-1}$ predicts. Zero deviations across the whole battery (levels 20..100).

2 Consequences

One surface. The critical surface $u = 6\rho^2 m$ is the first knife's line: at leading order there is only one boundary in the entire trajectory sector, and every higher constraint is a power of it. The $(j-1)$ -fold root splits at order $\sqrt{m}$ into $j-1$ simple roots — $\lfloor j/2 \rfloor$ finite negativity bands (odd j : bands only; even j : bands plus the half-line on the far side, which coincides with the first knife's own dead side).

The tangency explained. Minimizing $6\rho^2 m - 4m$ over the level at fixed λ gives $12\rho - 6\rho^2 \cdot \rho^{-1} \dots$ — the stationary ratio $\rho^* = 1 + 1/\sqrt{3}$ and the envelope asymptote $D = (12 + 4\sqrt{3})\lambda$ of [1]. The exact tangency that made the blade theorem [3] delicate is thus the $j=3$ instance of a universal fact: *every* knife's asymptotic cone is tangent to the envelope asymptote, because every cone is a power of the one whose envelope it is.

The road to full completeness. The collapse organizes the large- m tail of the completeness conjecture uniformly in j : one lemma family (containment of bands near the surface, plus the tangency structure) replaces infinitely many per- j cones. What remains hard, and open, is the $\sqrt{m}$ -order control of band edges near the optimal level — the regime where the blade theorem's margins (≈ 2.15 dimensions) live. At moderate m the asymptotic surface is quantitatively far from the exact laws (e.g. level 8 at $\lambda = 5$: 152.8 asymptotic vs 94 exact), so certificates remain irreplaceable there.

3 Verification

All numbers in exact rational arithmetic. The zone battery (`collapse_zones.py`): knives $j = 2..8$, levels $n = 20..100$, every negativity zone computed exactly and compared with the collapse prediction — band counts $\lfloor j/2 \rfloor$ and clustering near the critical line: zero deviations. The earlier completeness sweep of [2] (3,053,832 exact verdicts, all knives, zero violations) and the blade theorem [3] are both consistent with, and now organized by, the collapse picture.

Explain it to anyone

In our previous papers the danger zones above the shore looked like a fleet of unrelated blades — one shape for the second knife, another for the third, on and on forever. This paper finds the secret: they are not unrelated at all. *Every knife is a power of the first one.* Square the first knife and you get the second’s shape; cube it, the third’s. That is why all their danger zones huddle around a single line, and why the fleet sails tangent to the shore: there is really only one boundary in this world, seen through different exponents. Knowing the family secret, we can now hunt the final theorem with one key instead of infinitely many.

Reproducibility and AI disclosure

The exact term-ratio derivation, the zone battery and all artifacts are released with this paper. This research was conducted by the author working with an AI research assistant (Claude, Anthropic). A tempting shortcut — positive-definiteness of the residue root polynomial plus the Schoenberg product theorem, which would have proven full completeness in one stroke — was tested and found *false* (the root polynomial has negative Gegenbauer coefficients inside the allowed region); the negative result is preserved in the public log.

References

- [1] A. Pluzhnik, “The Shore of Closed-String Gravity,” DOI 10.5281/zenodo.21944818.
- [2] A. Pluzhnik, “A Master Positivity Formula for the Cheung–Hillman–Remmen Graviton Family,” DOI 10.5281/zenodo.21947272.
- [3] A. Pluzhnik, “The Blades Never Touch the Shore,” DOI 10.5281/zenodo.21948833.